

D) all of the above

3- Definition for an array arr of 8 pointers that point to variables of type float is

- A) *float arr[8];
- B) float* arr[8];
- C) float pointer[8];
- D) int *arr[8];

4- Under what conditions a destructor destroys an object?

- A) Scope of existence has finished
- B) Object dynamically assigned and it is released using the operator delete
- C) Program terminated
- D) All of the above

Question 4: (5 points) Given the following namespace answer the following?

using namespace std;

namespace hhh{

int * fun1(int *p){
*p+=1;
return p;}

int & fun2(int &c){
++c;
return c; }

void fun3(int k){
k+=2;
cout<<k;
return; }

namespace ddd{

struct mystruct
{ string g="abc";
int x=9;}xm; }

class myc{ public: int x,y;};

1-
int x=7;
int*q=hhh::fun1(&x);
cout<<x; // 8
cout<<*q; // 8

2-
int z=5;
int &g=hhh::fun2(z);
cout<<z; 6
cout<<g; 6

3-
int k=5;
hhh::fun3(k); 75
cout<<k; 5

4-
hhh::ddd::mystruct xk;
cout<<xk.g; abc
cout<<hhh::ddd::xm.x; 9

5-
using namespace hhh;
myc xc;
xc.x=2; 12
xc.y=1;
cout<<xc.x+xc.y+ 2+1+9=12
hhh::ddd::xm.x;

Question 1:

1- A ✓ 2- D ✓ 3- C ✓ 4- B ✓ 5- B ✓ 6- B ✓ 7- A ✓

Question 2:

1- B ✓ 2- D ✓ 3- A ✓ 4- C ✓

Question 3:

1- A ✗ 2- C ✗ 3- B ✓ 4- D ✓

Question 4:

1- 88 2- 66 3- 75 4- abc9 5- 12

Question 1:(7 points) Given the following code Answer the following

```
class TypeA{
private:
    int x,*p; char c;
public:
    TypeA(int a)
    { x = a; }
    TypeA(int *q)
    { p = q; }
    TypeA(char c)
    { c = c; }
    void fun1(int a)
    { cout<<x+a; }
    int * fun2(int *k)
    {
        *k+=*p;
        return k; }
    char & fun3(char m)
    {
        char *g=new char(m);
        return *g; }
    int fun4(TypeA k)
    { return pow(*(k.p),2); }
    char & fun4(char &h)
    {
        if(h=='a')
            return ++h;
        else
            return --h; }
};
```

1- output if these instructions were executed in main

```
TypeA xa(1),xb(new int(7)), xc('c');
int x=2;
int *p=&x;
p=xb.fun2(new int(2));
char cc='b';
cout<<xa.fun4(cc);
```

A) a

B) b

C) cc

D) other

2- How many overloaded constructor in the class TypeA?

A) 4

B) 5

C) none

D) other

3- which function returns a reference

A) fun1

B) fun2

C) fun3

D) other

4- Which function returns a pointer?

A) fun1

	☒ B) fun2 ☐ C) fun3 ☐ D) other
5- output if these instructions were executed in main TypeA xa(1),xb(new int(7)), xc('c'); cout<<xa.fun4(xb);	☐ A) 16 ☒ B) 49 ☐ C) cc ☐ D) other
6- Output if these instructions were executed in main? TypeA xa(1),xb(new int(7)), xc('c'); int x=2; int *p=&x; p=xb.fun2(new int(2)); cout<<xc.fun3('d');	☐ A) c ☒ B) d ☐ C) e ☐ D) other
7- what is the output if these instructions were executed in main TypeA xa(1),xb(new int(7)), xc('c'); int x=2; int *p=&x; p=xb.fun2(new int(2)); cout<<*p;	☒ A) 9 ☐ B) d ☐ C) 7 ☐ D) other
Question 2: (4 points) What will happen in the following C++ code snippet?	
1- int a =100, b =200; int *p = &a, *q = &b ; p = q;	☐ A) b is assigned to a ☒ B) p now points to b ☐ C) a is assigned to b ☐ D) q now points to a
2- int a = 5, b = 10, c = 15; int *arr[] = {&a, &b, &c}; cout << arr[1];	☐ A) 5 ☐ B) 10 ☐ C) 15 ☒ D) it will return some random number
3- char *ptr; char str[] = "abcdefg"; ptr = str; ptr += 5; cout << ptr;	☒ A) fg ☐ B) cdef ☐ C) defg ☐ D) abcd
4- class test{ int x,y;}; main() {test t; t.x=1; t.y=t.x++; cout<<t.x<<t.y;}	☐ A) 11 ☒ B) 12 ☒ C) error ☐ D) none
Question 3: (4 points) choose the correct answer	
1- A pointer is	☒ A) the address of a variable ☐ B) an indication of the variable to be accessed next ☐ C) a variable for storing addresses ☐ D) the data type of an address variable
2- *test	☐ A) refers to the contents of test ☐ B) dereferences test ☐ C) refers to the value of the variable pointed to by test